

ALEXANDER REEVES

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EDUCATION

PhD, ETH Zürich

September 2021 - May 2026 (expected)

PhD candidate in Professor Alexandre Refregier's Cosmology group with secondary supervision from Professor Andrina Nicola (University of Manchester)

Multiprobe analysis combining Cosmic Microwave Background and Large Scale Structure data

Using the Effective Field Theory of Large Scale Structure (EFTofLSS) to analyse galaxy clustering

Masters, Downing College, University of Cambridge

September 2020 - July 2021

Masters: Part III Natural sciences (Astrophysics)

Masters grade: 1st class (ranked 4th in cohort)

Master thesis supervised by Dr. Sunny Vagnozzi, Professor Blake Sherwin, and Professor George Efstathiou

Bachelors, Downing College, University of Cambridge

September 2017 - July 2020

Undergraduate: Part II Physics

Undergraduate grade: 1st class

AWARDS

Doc.Mobility fellowship, ETH Zürich

2024

Awarded (9100CHF) fellowship to undertake research at the University of California, Berkeley and Lawrence Berkeley National lab under the supervision of Dr. Simone Ferraro and Professor Martin White.

1st place: International PLANCKS theoretical Physics competition

2020

Part of the 4-person University of Cambridge team that came first out of 45 University teams at the international finals.

Whitelegg and Unwin Scholarships, Downing College

2018 & 2021

Scholarships for academic achievement.

STUDENT SUPERVISION

Domitille Chebat, Master thesis co-supervisor

February 2024 - September 2024

Thesis title: 'Realistic field level simulations with the effective field Theory of Large Scale Structure'

Laura Reymond, Master thesis co-supervisor

February 2024 - September 2024

Thesis title: 'Precise computation of the angular power spectrum at large angular scales'

Laura Reymond, Semester thesis supervisor

September 2023 - December 2023

Thesis title: 'Precise computation of the angular power spectrum at large angular scales'

Fabio Holdener, Master thesis co-supervisor

February 2023 - August 2023

Thesis title: 'Weak Lensing With 21cm Intensity Mapping'

Luca Morf, Master thesis co-supervisor

September 2022 - March 2023

Thesis title: 'Field-level Simulations from the Effective Field Theory of Large-Scale Structures'

Peter Bajzath, Semester project supervisor

September 2022 - December 2022

Thesis title: 'Investigating measurement tensions with Machine Learning emulators'

Luis Machado Poletti Valle, Semester project supervisor

February 2022 - May 2022

Thesis title: 'The Integrated Sachs-Wolfe effect on the lightcone'

RELEVANT TEACHING EXPERIENCE

Cosmology in group project (designed and supervised project)	<i>September 2025 - Present</i>
Teaching assistant for ‘Astrophysics II’	<i>February 2024 - September 2024</i>
Teaching assistant for ‘Cosmological probes’	<i>September 2023 - December 2023</i>
Teaching assistant for ‘Theoretical cosmology’	<i>February 2023 - May 2023</i>

SELECTED TALKS AND POSTERS

Invited talk, Tensions in Cosmology Invited conference talk based on arXiv:2502.01722	<i>Corfu, September 2025</i>
Invited talk, Recontres du Vietnam Invited conference talk based on arXiv:2502.01722	<i>Quy Nhon, August 2025</i>
INPA colloquium, Lawrence Berkeley National Laboratory Colloquium talk based on arXiv:2502.01722	<i>Berkeley, February 2025</i>
Tea leaks talk at Stanford University Short talk based on arXiv:2502.01722	<i>Stanford, January 2025</i>
Poster at ‘Future Science with CMB x LSS’ Delivered a fireside talk as well as a poster based on arXiv:2309.03258	<i>Kyoto, March 2023</i>
Tensions in Cosmology Delivered a 10-minute talk on results of the paper: arXiv:2207.01501	<i>Corfu, August 2022</i>
Talk at NBI Summer school: ‘Neutrinos here there and everywhere’ Delivered a 10-minute talk on results of the paper: arXiv:2207.01501: link	<i>Copenhagen, June 2022</i>

COLLABORATION MEMBERSHIPS

Dark Energy Spectroscopic Instrument (DESI) member
Legacy Survey of Space and Time-Dark Energy Science Collaboration (LSST-DESC) member

SOFTWARE

1. Led the development of `UFalcon2` a software tool in `Python` for rapidly post-processing N-body code output into signal maps for several cosmological probes source code, documentation.
2. Led the development of `PyBird-JAX` a software tool in `Python-JAX` for rapid, differentiable EFTofLSS predictions source code.

PUBLICATIONS

1. **A.Reeves**, S.Ferraro, A.Nicola, A.Refregier *Multiprobe constraints on early and late time dark energy*, September 2025, submitted to Physical Review D, arXiv:2510.06114
Contribution: conceived and led the project, designed methodology, performed analysis, wrote the manuscript.
2. **A.Reeves**, P. Zhang, H. Zheng, *Debiasing inference in large-scale structure with non-flat volume measures*, July 2025, submitted to JCAP, arXiv:2507.20990
Contribution: co-led the project, computational analyses, co-wrote the manuscript.
3. **A.Reeves**, P. Zhang, H. Zheng, *PyBird-JAX: Accelerated inference in large-scale structure with model-independent emulation of one-loop galaxy power spectra*, July 2025, submitted to JCAP, arXiv:2507.20990
Contribution: led the project; built and ran the computational pipeline, paper writing.

4. L.Reymond, **A.Reeves**, P.Zhang, A.Refregier, *SwiftC_ℓ fast differentiable angular power spectra beyond Limber*, May 2025, accepted in JCAP, arXiv:2504.01669
Contribution: concept, supervision, paper editing.
5. Di Valentino et al. (including **A.Reeves**), *The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics*, April 2025, arXiv:2504.01669
Contribution: wrote part of the 'Early dark energy and variants' section.
6. **A.Reeves**, A.Nicola, A.Refregier, *Tuning the cosmic instrument: robust cosmology through combined probes*, February 2025, accepted in JCAP, arXiv:2502.01722
Contribution: led the project; conducted the computational analysis, wrote the majority of the manuscript.
7. **A.Reeves**, A.Nicola, A.Refregier, T.Kacprzak, L.F.Machado Poletti Valle, *12×2pt combined probes: pipeline, neutrino mass, and data compression*, January 2024, accepted in JCAP, arXiv:2309.03258.
Contribution: led the project, performed the majority of the computational analysis and writing of the paper.
8. **A.Reeves**, L.Herold, S.Vagnozzi, B.Sherwin, E.Ferreira, *Restoring concordance with Early Dark Energy and Massive neutrinos*, April 2023, accepted in MNRAS, arXiv:2207.01501.
Contribution: project lead, carried out the bulk of theoretical and computational work, wrote the manuscript.